Artificial Intelligence and Machine Learning

Module 1 — Introduction to Artificial Intelligence

- Definition and scope of Artificial Intelligence
- Historical overview and milestones in AI
- AI applications in various fields
- Agents and environments

Module 2 — Knowledge Representation and Reasoning

- Propositional logic
- First-order logic and predicate calculus
- Knowledge representation using logic
- Inference rules and resolution

Module 3 — Introduction to Statistical Learning

- Overview of statistical learning and its applications
- Supervised vs. unsupervised learning
- Bias-variance trade-off
- Performance evaluation and cross-validation

Module 4 — Linear Regression

- Simple linear regression
- Multiple linear regression
- Model assessment and selection
- Polynomial regression and model flexibility
- ◆ Lab: Python / R

Module 5 — Classification

- Logistic regression
- Linear discriminant analysis (LDA)
- K-nearest neighbors (KNN) classification
- Model evaluation and validation techniques
- ◆ Lab: Python / R

Module 6 — Introduction to Machine Learning

- Overview of different machine learning algorithms
- Decision Trees and Random Forests
- Support Vector Machines (SVM)
- Ensemble methods (bagging and boosting)
- ◆ Lab: Python / R

Module 7 — Unsupervised Learning

- K-means clustering
- Hierarchical clustering
- Association rules and market basket analysis
- Anomaly detection
- ◆ Lab: Python / R

Module 8 — Correlation & Simple Linear Regression

- Bivariate data: scatter diagram, correlation & rank correlation
- Theory of attributes, independence and association

• Assumptions of linear regression • Residual analysis and model diagnostics ◆ Lab: Python / R	• Estimating regression coefficients